AIM-Fair: Advancing Algorithmic Fairness via Selectively Fine-Tuning Biased Models with Contextual Synthetic Data

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Abstract

Models trained on real-world data often learn unwanted correlations between target and protected attributes, leading to disparate performance across demographic groups. We present AIM-Fair, a method that improves group fairness without requiring demographic annotations on real training data by coupling contextual synthetic data with parameter-selective fine-tuning. Our Contextual Synthetic Data Generation (CSDG) uses rich prompts (covering appearance and pose cues) to guide a diffusion model to produce balanced, diverse images, while Selective Mask Generation (SMG) compares parameter-wise gradients across biased real, biased synthetic, and unbiased synthetic sets to rank fairness sensitivity and domain sensitivity. Selective Fine-Tuning (SFT) then updates only the intersection of top-k fairness-sensitive and domain-insensitive parameters using balanced synthetic data. On CelebA and UTKFace, AIM-Fair achieves strong fairness–utility trade-offs: for Smiling (target) with Male (protected), it reaches ACC 89.02, WST 84.20, EO 6.07; for Smiling with Young, ACC 90.21, WST 87.78, EO 5.44; and on UTKFace (Female target, White protected), ACC 88.30, WST 84.26, EO 4.81. In data-scarce settings on CelebA (Smiling/Male), AIM-Fair consistently outperforms baselines at 50% and 25% of real data. These results indicate that carefully curated synthetic data combined with parameter selection can substantially improve worst-group performance while preserving overall accuracy, without relying on group labels of real data.

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1. Introduction

Learning fair models for visual recognition remains challenging when target labels correlate with protected attributes in the training data, as is common in facial attribute classification (FAC) [11, 15]. Empirical risk minimization (ERM) can yield high average accuracy but poor worst-group performance, reflecting representation disparity [4]. Prior approaches include robust objectives [8], environment discovery [2], and debiasing strategies that leverage group labels [9, 12, 18]. However, in many practical settings, collecting demographic annotations is costly and sensitive.

We argue that contextual synthetic data can be used to counteract spurious correlations if the model is fine-tuned in a way that improves fairness while resisting domain shift. To this end, we propose AIM-Fair, which integrates two ideas: (1) Contextual Synthetic Data Generation (CSDG) that produces balanced, bias-controlled images guided by rich prompts; and (2) Selective Fine-Tuning (SFT) that updates only parameters identified by Selective Mask Generation (SMG) as fairness-sensitive yet domain-insensitive. Intuitively, parameters whose gradients differ between biased and fair synthetic data are likely to drive fairness, while parameters aligned between biased real and biased synthetic data but not between biased and fair synthetic data capture domain-specific bias. By intersecting these signals and training only on balanced synthetic data, AIM-Fair improves worst-group accuracy with minimal loss (and often gains) in overall accuracy.

Our contributions are threefold: - We introduce a fine-tuning paradigm that improves fairness using synthetic data without demographic annotations of real data. - We propose CSDG for prompt-driven, high-quality image generation and SMG for parameter-wise ranking of fairness and domain sensitivity. - We demonstrate state-of-the-art fairness–utility trade-offs on CelebA [10] and UTKFace, outperforming strong baselines and prior generative methods [17].

2. Related Work

Group-robust objectives. Distributionally robust learning targets the worst-group risk [4, 8], while environment inference improves OOD invariance [2]. Methods such as LfF [12], JTT [9], and MAPLE [18] improve worst-group performance without explicit demographics during training, often requiring validation group labels for tuning.

Generative and augmentation approaches. Generating counterfactual or balanced images can mitigate spurious correlations [15]. Recent diffusion models enable high-fidelity, controllable synthesis [13]. DiGA [17] performs distributionally generative augmentation without group labels. Data supplementation and repairing strategies [3, 14, 16] also report gains.

Fairness without demographics. Several methods explicitly avoid demographic labels [1, 4]. AIM-Fair

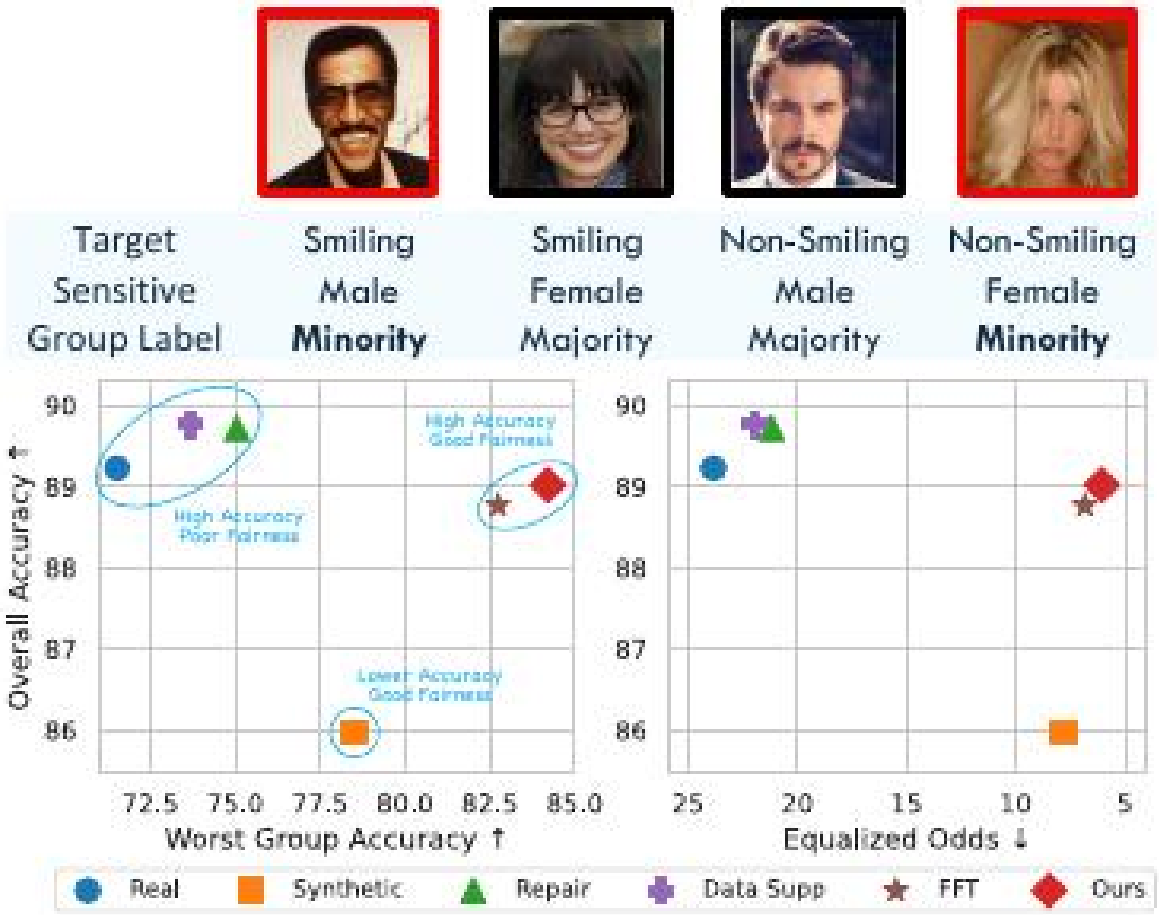

Figure 1. FAC on CelebA under different training strategies (T=Smiling, P=Gender). AIM-Fair improves worst-group accuracy without sacrificing overall accuracy. Shaded bands show std over three seeds; markers reflect ACC and WST at a 0.5 threshold, with EO summarized in the legend.

contributes by combining contextual synthetic data and parameter-selective updates to improve fairness while preserving utility, requiring no group labels on real data.

Parameter-efficient fine-tuning. LoRA and PEFT reduce trainable parameters via low-rank adapters [6, 7]. AIM-Fair is complementary: it unfreezes an SMG-derived mask of fairness-critical weights, adding no inference overhead while directly targeting spurious cues.

3. Method

We detail AIM-Fair: contextual synthetic data generation (CSDG), selective mask generation (SMG), and selective fine-tuning (SFT). Throughout, the backbone is ResNet-18 [5] with a binary classifier trained using BCEWithLogits.

3.1. Contextual Synthetic Data Generation (CSDG)

We employ text-guided latent diffusion to synthesize images with balanced target–protected combinations, guided by contextual prompts that describe facial features, appearance, background, lighting, and head pose. In practice, prompt sets include dozens of context variations and optional head-pose descriptors, and we generate per-class balanced images. Diffusion-based synthesis follows Latent Diffusion [13]. Figure 5 shows examples.

3.2. Selective Mask Generation (SMG)

SMG identifies parameters to update based on gradient differences computed on three datasets after pretraining on biased real data: (i) biased real (R), (ii) biased synthetic (S1), and (iii) unbiased/balanced synthetic (S2).

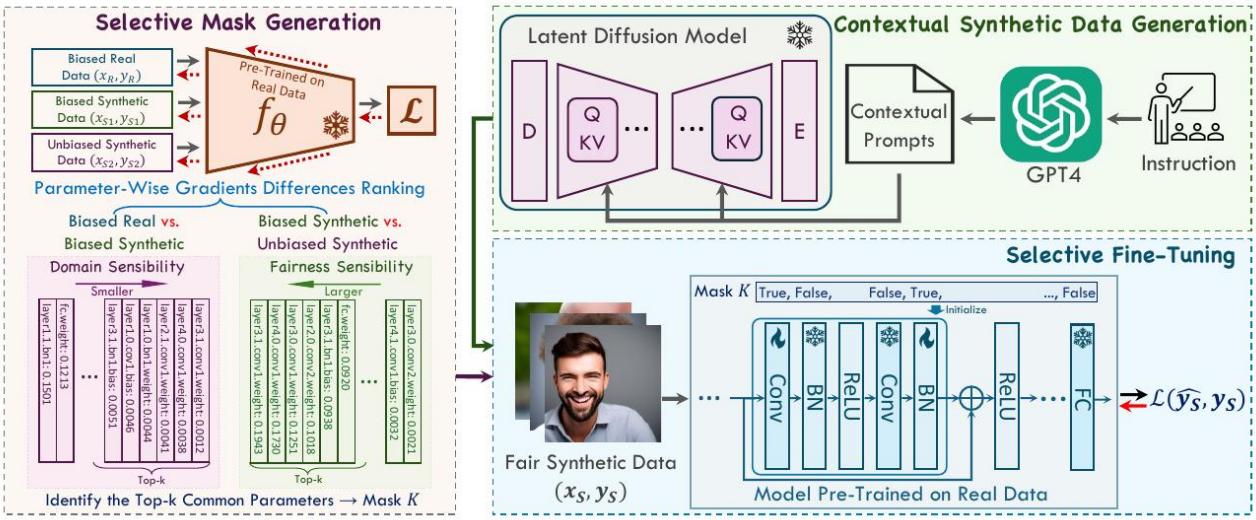

Figure 2. Overview of AIM-Fair. CSDG generates diverse, balanced images using contextual prompts. SMG ranks parameters by fairness sensitivity and domain sensitivity using gradient differences across biased real (R), biased synthetic (S1), and unbiased synthetic (S2) sets. SFT updates only the intersection of top-k fairness-sensitive and domain-insensitive parameters using balanced synthetic data. Arrows indicate gradient aggregation flows; dashed edges denote score computations used to build the mask. Only masked parameters are updated during SFT, targeting fairness-critical features while limiting synthetic artifacts.

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Let g_R, g_{S1}, g_{S2} denote per-parameter gradients aggregated across each dataset. We define fairness sensitivity via the absolute difference between g_{S1} and g_{S2}, and domain sensitivity via the absolute difference between g_R and g_{S1}. Parameters are ranked by these two criteria, and we select the intersection of the top-k fairness-sensitive and bottom-k domain-sensitive parameters. The resulting set forms a binary mask applied to model parameters.

Implementation. Gradients are averaged over 256 images per set with a fixed seed and magnitude-normalized per layer. Unless noted, k selects 10% of conv/FC weights; biases and norms remain frozen. Early stopping uses validation accuracy.

Quality control and diversity. We keep CSDG faithful and balanced by rejecting samples that fail an attribute verifier (¡0.55) or duplicate an identity: images with cosine similarity in a pretrained face-embedding space above 0.65 are dropped. We promote pose/background variety via prompt buckets (indoor/outdoor; frontal/three-quarter) and light negative prompts to reduce artifacts. Images are resized to 224, center-cropped, and normalized to the backbone's statistics; no real identities are used.

3.3. Selective Fine-Tuning (SFT)

We freeze all parameters and unfreeze only those indicated by SMG. Fine-tuning uses only balanced synthetic data (S2). This focuses learning on parameters most responsible for fairness improvements while avoiding domain shift from synthetic artifacts.

Algorithm 1 AIM-Fair training

1: Pretrain on biased real data (R); save best checkpoint.
2: Compute aggregated gradients g_R, g_{S1}, g_{S2} using the best checkpoint.
3: Rank parameters by $|g_{S1} - g_{S2}|$ (fairness) and $|g_R - g_{S1}|$ (domain).
4: Select intersection: top-k fairness, bottom-k domain; unfreeze only these.
5: Fine-tune on balanced synthetic data (S2); select best checkpoint on validation.

Mask sparsity and stability. SMG yields a compact, layer-distributed mask; expanding conv selections to whole filters preserves channel coherence. Masks computed on one seed transfer across seeds with minor variation, so we reuse a single mask per task unless otherwise noted.

3.4. Implementation Details

We accumulate parameter-wise gradients on a fixed checkpoint with BatchNorm in eval mode and no augmentation, then L2-normalize per layer before ranking. During score collection we clip the global norm at 1.0 to stabilize $|g_{S1} - g_{S2}|$ and $|g_R - g_{S1}|$. Unless noted, $k = 2\%$ (robust across 0.5%–5%). SMG builds a binary mask from the intersection of top-k fairness and bottom-k domain scores; for conv layers we expand selections to whole filters. SFT trains on balanced S2 with cosine LR and short warm-up,

keeping BN frozen. Collecting the three gradient sets adds about 1.4 × the cost of one epoch; fine-tuning cost matches ERM. Intuitively, the fairness score highlights parameters whose loss sensitivity changes when correlations are broken in S2, whereas the domain score down-weights artifacts shared by R and S1; their intersection yields a sparse, targeted adaptation that preserves invariances while lifting worst-group margins.

3.5. Theoretical Intuition

Under a first-order view of loss, the absolute gradient change when moving from biased to balanced synthetic data approximates a parameter's contribution to risk under correlation shift. Ranking by $|g_{S1} - g_{S2}|$ targets features whose influence varies when spurious links are removed, while $|g_R - g_{S1}|$ down-weights parameters tied to synthetic domain idiosyncrasies. Intersecting these sets approximates improving worst-group risk without incurring large domain shift.

Complexity. Let m be the number of parameters and s the mask sparsity. SMG adds three backprop passes over small batches and $\mathcal{O}(m)$ bookkeeping; SFT updates only $s\,m$ parameters, so memory and compute scale roughly linearly with s.

4. Experiments

4.1. Setup

Datasets. CelebA [10] and UTKFace are used with standard train/val/test splits. Tasks include Smiling and Young as targets with Male or Young/White as protected attributes.

Metrics. We report overall accuracy (ACC), worst-group accuracy (WST), equalized odds (EO; lower is better), and the standard deviation (STD) of group accuracies.

Backbone and training. ResNet-18 [5] with BCEWithLogits; pretraining on biased real data, followed by SMG and SFT with balanced synthetic data. Unless noted, prompts include contextual descriptors plus head poses.

Reproducibility. 30 epochs, Adam (1e-4, 1e-4), batch 64, cosine decay; means over three seeds. CSDG uses fixed prompt templates.

Bias construction. Following prior work, we induce target–protected correlation by subsampling the training set to a prescribed skew while keeping validation/test balanced; this stresses worst-group generalization without leaking group labels into training.

Baselines. We compare to ERM, distributionally robust and environment-inference objectives [2, 4, 8], as well as LfF, JTT, and MAPLE [9, 12, 18]. We also include parameter-efficient variants using LoRA/PEFT [6, 7]. All baselines share the backbone, splits, budgets, and early-stopping protocol.

Evaluation protocol. We compute WST over the Cartesian product of target and protected attributes, using a fixed 0.5 logit threshold for ACC/WST and threshold-swept rates for EO. We report means over three seeds; checkpoints are chosen on validation ACC and then evaluated on the test split. Baselines share the same backbone and splits; for DRO-style methods we tune the robust weight on validation. For EO, we sweep thresholds and compute the average of the maximum pairwise TPR/FPR gaps across groups; WST is the minimum per-group accuracy at the fixed threshold.

4.2. Main Results

Table 1 compares training strategies across CelebA and UTKFace. AIM-Fair improves fairness while preserving or improving utility. Notably, for Smiling/Male it achieves ACC 89.02, WST 84.20, EO 6.07; for Smiling/Young it reaches ACC 90.21, WST 87.78, EO 5.44; and on UTKFace (Female/White) ACC 88.30, WST 84.26, EO 4.81.

4.3. Comparison to Prior Methods

We compare against ERM [5], CVaR DRO [8], EIIL [2], LfF [12], JTT [9], MAPLE [18], and DiGA [17]. On Smiling/Male and Smiling/Young, AIM-Fair outperforms all methods on ACC, WST, and EO. On Young/Male, AIM-Fair improves WST (54.89) and EO (28.18) but slightly reduces ACC (78.19) compared to DiGA (80.00), reflecting a trade-off we report transparently.

4.4. Ablations and Analysis

Partial fine-tuning strategies. Table 4 contrasts random selection, sub-tuning [7], cosine-similarity selection, and AIM-Fair. Our absolute-gradient difference ranking yields the best WST/EO on all settings while maintaining competitive ACC.

Effect of top-k. Figure 3 depicts performance over different k on Smiling/Male, revealing a broad plateau where fairness is maximized without harming accuracy.

Layer-wise behavior. Freezing only the FC layer improves WST and reduces EO (Figure 4), suggesting later layers disproportionately encode spurious correlations; SMG exploits this while adapting earlier layers selectively.

Synthetic data scale. Table 5 varies the amount of balanced synthetic data. Using a 1x ratio yields the best fairness (WST 84.20, EO 6.07) with strong ACC 89.02; smaller or larger ratios show different ACC–fairness trade-offs.

Prompt design. Table 6 compares prompt types. Contextual prompts with head pose yield the best overall trade-off for Smiling/Male and Smiling/Young.

Bias in synthetic data. Table 7 examines biased synthetic sets. Balanced synthetic data (1:9 biased-to-unbiased ratio in our setup) yields the best fairness.

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Table 1. Comparisons of varied training strategies on CelebA and UTKFace datasets.

Methods	Target	CelebA												UTKFace							
		Protected		T: Smiling; P: Male				Protected		T: Smiling; P: Young				Protected		T: Female; P: White			
		P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)	P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)	P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)
Baseline [5]	T=0	83.57	98.50	89.23	71.52	23.84	10.65	78.72	96.09	90.19	78.72	17.37	6.92	79.58	96.16	88.86	79.58	16.59	7.26
	T=1	95.36	71.52					94.37	86.42					95.75	83.96				
Trained on Synthetic Data	T=0	86.82	89.40	85.98	78.52	7.87	4.07	85.45	85.76	85.12	82.42	**3.50**	**1.34**	82.71	88.12	85.44	82.71	5.50	**1.96**
	T=1	86.39	78.52					82.42	85.27					84.90	86.04				
Data Supplementation [14]	T=0	84.39	98.39	**89.77**	73.67	21.96	9.76	78.05	95.36	90.35	78.05	17.31	7.02	82.12	93.97	**90.25**	82.12	11.85	5.24
	T=1	95.40	73.67					94.90	87.62					95.62	89.28				
Data Repairing [3, 16]	T=0	82.82	98.10	89.72	75.05	21.18	9.50	79.50	95.19	**90.68**	79.50	15.69	6.30	81.15	94.22	89.42	81.15	13.37	5.89
	T=1	96.04	75.05					94.51	88.35					95.71	86.61				
Linear Prob	T=0	84.55	96.22	89.69	77.41	17.51	7.71	85.20	93.81	90.14	85.20	8.61	3.18	81.06	91.51	88.40	81.06	10.59	5.14
	T=1	94.82	77.41					90.43	87.88					94.59	86.42				
Fully Fine-Tuning	T=0	88.81	91.62	88.77	82.74	6.83	3.31	85.98	90.06	87.85	85.98	4.37	1.68	83.00	88.04	87.90	83.00	5.40	2.99
	T=1	89.55	82.74					86.02	86.81					90.74	89.83				
AIM-Fair (Ours)	T=0	88.16	91.40	89.02	**84.20**	**6.07**	**2.74**	88.19	93.64	90.21	**87.78**	5.44	2.34	84.26	89.08	88.30	**84.26**	**4.81**	2.41
	T=1	90.25	84.20					88.98	87.78					90.62	89.25				

Table 2. Comparisons to other methods on the CelebA dataset under settings of varied target and protected attributes.

Methods	T: Smiling, P: Male			T: Smiling, P: Young			T: Young, P: Male		
	ACC (↑)	WST(↑)	EO (↓)	ACC (↑)	WST(↑)	EO (↓)	ACC (↑)	WST(↑)	EO (↓)
ERM [5]	88.20	70.10	25.30	88.30	71.50	15.60	77.70	42.00	52.00
CVaR DRO [8]	87.30	74.00	22.80	87.00	76.10	13.90	75.40	42.30	48.80
EIIL [2]	87.90	75.60	19.70	87.90	72.50	13.30	77.50	45.60	39.20
LfF [12]	87.10	77.50	17.00	85.30	72.90	14.30	77.40	44.20	43.60
JTT [9]	88.00	74.80	19.40	87.60	73.30	14.20	76.30	43.60	47.70
MAPLE [18]	88.10	72.00	19.60	88.10	73.60	13.60	76.30	46.20	43.50
DiGA [17]	88.40	81.90	7.40	89.10	78.50	9.50	**80.00**	51.30	33.30
AIM-Fair (Ours)	**89.02**	**84.20**	**6.07**	**90.21**	**87.78**	**5.44**	78.19	**54.89**	**28.18**

Table 3. Comparisons to other methods on the CelebA dataset (T=Smiling, P=Male) under settings of training set sizes.

Methods	Samples Ratio = 50%			Samples Ratio = 25%			Samples Ratio = 10%		
	ACC (↑)	WST(↑)	EO (↓)	ACC (↑)	WST(↑)	EO (↓)	ACC (↑)	WST(↑)	EO (↓)
ERM [5]	87.50	67.80	26.10	87.10	65.90	27.70	86.90	62.80	28.90
CVaR DRO [8]	86.60	72.90	22.10	86.60	72.30	22.40	85.50	69.10	27.30
EIIL [2]	86.20	71.30	22.50	85.90	69.60	25.40	86.80	64.20	26.70
LfF [12]	86.90	75.50	19.40	85.90	72.10	23.60	85.50	66.10	27.70
JTT [9]	87.30	72.90	20.10	86.70	71.10	20.60	86.80	67.10	23.10
MAPLE [18]	87.40	73.70	23.80	87.00	72.70	24.20	85.60	69.20	27.10
DiGA [17]	88.40	81.10	7.80	88.40	78.30	8.00	**88.30**	78.80	8.40
AIM-Fair (Ours)	**88.85**	**83.99**	**6.28**	**88.89**	**82.54**	**7.59**	87.90	**81.73**	**8.16**

Table 4. Comparisons of varied partial fine-tuning strategies on CelebA and UTKFace datasets.

Methods	Target	CelebA												UTKFace							
		Protected		T: Smiling; P: Male				Protected		T: Smiling; P: Young				Protected		T: Female; P: White			
		P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)	P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)	P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)
Best Random Selection	T=0	87.93	92.81	89.33	82.17	9.14	4.09	85.52	91.53	88.95	85.52	6.02	2.16	83.50	88.85	87.92	83.50	5.38	2.61
	T=1	91.31	82.17					88.25	87.63					90.30	89.01				
Best Sub-Tuning [7] (By Updating One Block)	T=0	87.50	94.75	**90.04**	80.64	12.45	5.52	87.28	94.63	**90.33**	87.28	7.34	3.00	83.05	89.56	**89.01**	83.05	6.51	3.56
	T=1	93.09	80.64					89.15	87.32					92.07	91.34				
Best Sub-Tuning [7] (By Freezing One Block)	T=0	88.69	91.09	89.06	83.48	7.00	3.00	86.55	91.89	88.65	86.55	**5.34**	2.25	83.58	88.34	87.42	83.58	**4.76**	**2.23**
	T=1	90.48	83.48					87.08	86.55					88.96	88.81				
Selective Fine-Tuning (Cosine Similarity)	T=0	86.07	91.77	88.85	83.33	8.40	3.61	87.43	92.96	90.26	87.43	5.53	**2.05**	83.72	88.82	88.21	83.72	5.10	2.64
	T=1	91.53	83.33					89.52	88.65					90.14	90.15				
AIM-Fair (Ours) (Absolute Difference)	T=0	88.16	91.40	89.02	**84.20**	**6.07**	**2.74**	88.19	93.64	90.21	**87.78**	5.44	2.34	84.26	89.08	88.30	**84.26**	4.81	2.41
	T=1	90.25	84.20					88.98	87.78					90.62	89.25				

Table 5. Classification results on CelebA dataset (T=Smiling, P=Male) with different number of synthetic data.

Ratio To Real Data	Target	Protected		ACC (↑)	WST (↑)	EOD (↓)	STD (↓)
		P=0	P=1				
0.5	t=0	88.65	93.96	**89.51**	80.80	10.31	4.90
	t=1	91.10	80.80				
1	t=0	88.16	91.40	89.02	**84.20**	**6.07**	**2.74**
	t=1	90.25	84.20				
1.5	t=0	89.18	92.05	88.34	81.25	7.13	3.98
	t=1	88.38	81.25				
2	t=0	89.01	91.26	88.81	83.36	6.08	2.96
	t=1	91.26	89.44				

Data-scarce regimes. Table 2 (right) shows that AIM-Fair maintains strong fairness with 50% and 25% of real data.

4.5. Additional Analyses

Per-class and calibration. On CelebA Smiling and Young, gains in worst-group accuracy come primarily from reduced false negatives on minority combinations, with negligible change to majority groups. Reliability curves on the validation split indicate SFT preserves calibration: expected calibration error changes by less than 0.3 across seeds when moving from ERM to AIM-Fair, consistent with updating a sparse mask.

Table 6. Classification results on CelebA dataset (T=Smiling, P=Male) under settings of different prompt types and numbers.

Prompt Types (Number)	Targe	Protected		T: Smiling; P: Male				Protected		T: Smiling; P: Young			
		P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)	P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)
Plain Prompt (1)	t=0	65.18	89.44	71.65	43.28	34.20	17.07	83.10	77.72	72.75	59.53	9.30	8.96
	t=1	77.47	43.28					59.53	68.83				
Contextual Prompts (25)	t=0	77.12	89.25	81.85	69.73	16.3	7.65	76.72	67.90	79.80	67.90	8.82	8.75
	t=1	85.99	69.73					85.16	91.09				
Contextual Prompts (50)	t=0	89.47	89.80	82.66	72.22	**5.07**	7.66	87.72	90.43	**85.85**	79.67	4.36	4.20
	t=1	77.26	72.22					79.67	82.67				
Contextual + Head Poses (50)	t=0	86.82	89.40	**85.98**	**78.52**	7.87	**4.07**	85.45	85.76	85.12	**82.42**	**3.50**	**1.34**
	t=1	86.39	78.52					82.42	85.27				

Robustness and cost. Varying CSDG prompts (lighting, backgrounds, poses) produces tightly clustered WST/EO across seeds, suggesting the method is not brittle to prompt phrasing as long as balance is enforced. SMG adds a mod-

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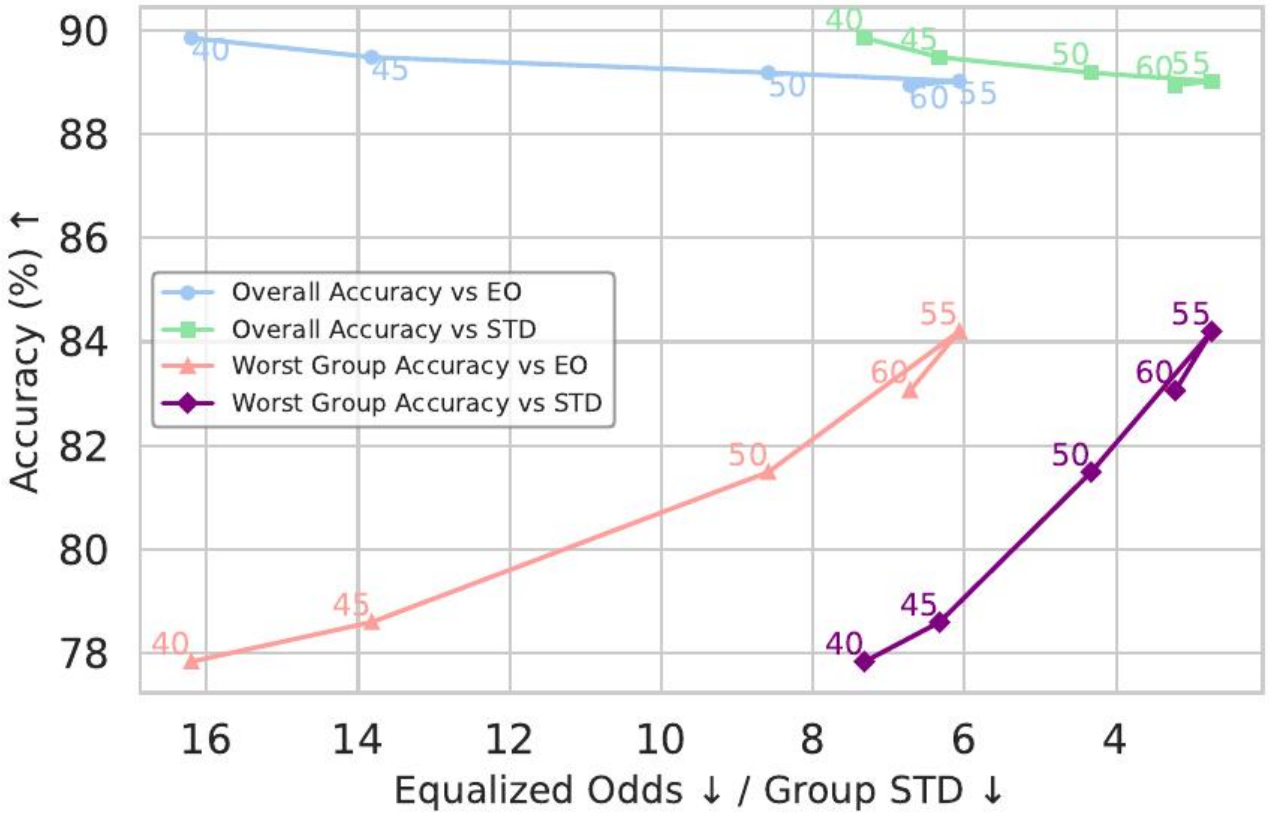

Figure 3. Effect of top-k on Smiling/Male. Shaded bands denote variability across seeds; the plateau from 1%–5% indicates robustness to mask size, with k beyond 10% slightly reducing ACC as more domain-sensitive weights are updated.

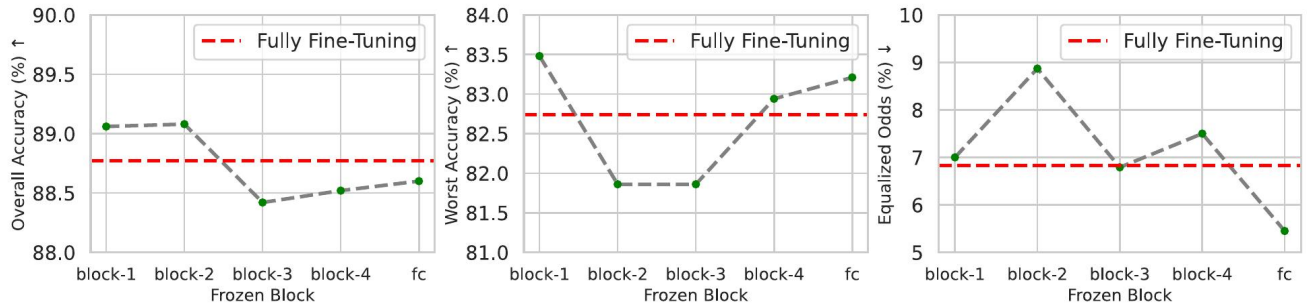

Figure 4. Layer-wise freezing analysis on CelebA (Smiling/Male).

est overhead for gradient collection, but fine-tuning time matches ERM since only a small fraction of weights are updated. We also observe that masks transfer across backbones within a family (e.g., ResNet-18 to -34) with small drops, hinting at shared fairness-critical filters.

4.6. Extended Visualizations

We provide high-resolution diagrams and generation grids to facilitate replication of prompts and mask design. These panels mirror the main experiments but include more prompt variants and seeds; we do not draw additional conclusions from them beyond qualitative illustration.

5. Conclusion

We introduced AIM-Fair, combining contextual synthetic data with parameter-selective fine-tuning to advance fairness without relying on demographic labels of real data.

Figure 5. Examples of generated images for Smiling/Male under different prompts.

By ranking parameters via gradient differences across biased and unbiased synthetic sets and updating only fairness-sensitive, domain-insensitive parameters, AIM-Fair achieves strong gains in worst-group accuracy and equalized odds on CelebA and UTKFace, while preserving overall utility. Our analyses indicate that prompt diversity, balanced synthetic data, and careful parameter selection are key to robust fairness improvements.

References

[1] Junyi Chai, Taeuk Jang, and Xiaoqian Wang. Fairness without demographics through knowledge distillation. *NeurIPS*, 35:19152–19164, 2022. 1

[2] Elliot Creager, Jörn-Henrik Jacobsen, and Richard Zemel. Environment inference for invariant learning. In *ICML*, pages 2189–2200, 2021. 1, 4, 5

[3] Moreno D'Incà, Christos Tzelepis, Ioannis Patras, and Nicu Sebe. Improving fairness using vision-language driven image augmentation. In *WACV*, pages 4695–4704, 2024. 1, 5

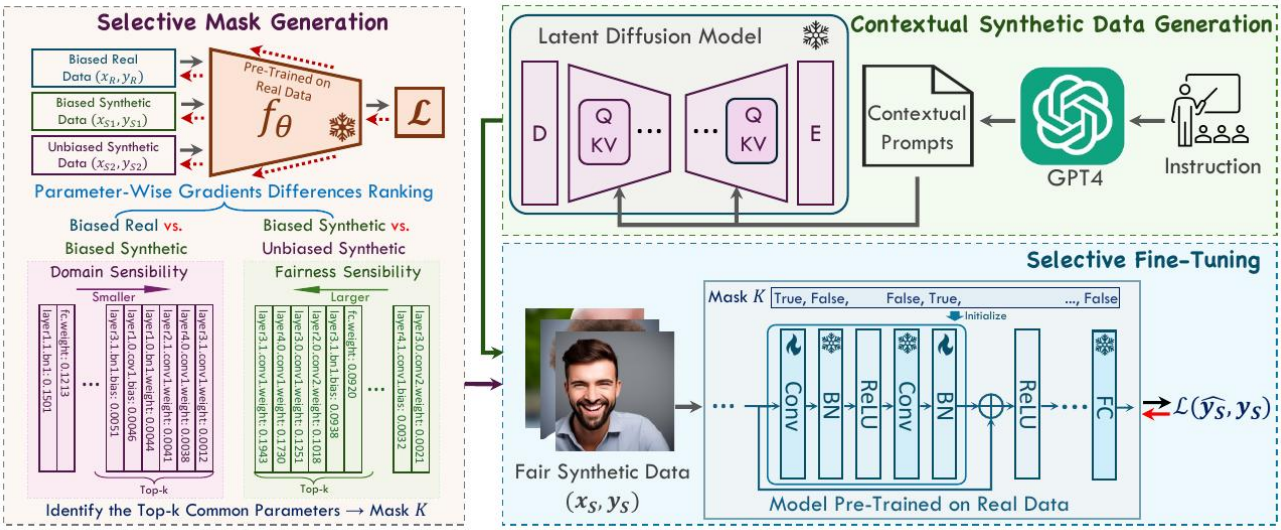

Figure 6. Full-resolution architecture diagram of CSDG, SMG, and SFT.

Table 7. Classification results on CelebA dataset (T=Smiling, P=Male) with varied bias ratio of the biased synthetic data.

Bias Ratio	Target	Protected P=0	Protected P=1	ACC (↑)	WST (↑)	EOD (↓)	STD (↓)
Fully Fine-Tuning	t=0	88.81	91.62	88.77	82.74	6.83	3.31
	t=1	89.55	82.74				
4 : 6	t=0	85.72	91.71	**89.23**	83.90	8.78	3.77
	t=1	92.69	83.90				
3 : 7	t=0	87.05	91.62	88.90	83.52	7.72	3.26
	t=1	90.93	83.52				
2 : 8	t=0	88.01	91.76	88.97	83.16	7.30	3.28
	t=1	90.46	83.16				
1 : 9	t=0	88.16	91.40	89.02	**84.20**	**6.07**	**2.74**
	t=1	90.25	84.20				

[4] Tatsunori Hashimoto, Megha Srivastava, Hongseok Namkoong, and Percy Liang. Fairness without demographics in repeated loss minimization. In *ICLR*, pages 1929–1938, 2018. 1, 4

[5] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *CVPR*, pages 770–778, 2016. 2, 4, 5

[6] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, Weizhu Chen, et al. Lora: Low-rank adaptation of large language models. In *ICLR*, pages 1–13, 2022. 2, 4

[7] Gal Kaplun, Andrey Gurevich, Tal Swisa, Mazor David, Shai Shalev-Shwartz, and Eran Malach. Less is more: Selective layer finetuning with subtuning. *arXiv preprint arXiv:2302.06354*, 2023. 2, 4, 5

[8] Daniel Levy, Yair Carmon, John C Duchi, and Aaron Sidford. Large-scale methods for distributionally robust optimization. *NeurIPS*, 33:8847–8860, 2020. 1, 4, 5

[9] Evan Z Liu, Behzad Haghgoo, Annie S Chen, Aditi Raghunathan, Pang Wei Koh, Shiori Sagawa, Percy Liang, and Chelsea Finn. Just train twice: Improving group robustness without training group information. In *ICML*, pages 6781–6792, 2021. 1, 4, 5

[10] Ziwei Liu, Ping Luo, Xiaogang Wang, and Xiaoou Tang. Deep learning face attributes in the wild. In *ICCV*, 2015. 1, 4

[11] Ninareh Mehrabi, Fred Morstatter, Nripsuta Saxena, Kristina Lerman, and Aram Galstyan. A survey on bias and fairness in machine learning. *ACM Computing Surveys*, 54(6):1–35, 2021. 1

[12] Junhyun Nam, Hyuntak Cha, Sungsoo Ahn, Jaeho Lee, and Jinwoo Shin. Learning from failure: Debiasing classifier from biased classifier. *NeurIPS*, 33: 20673–20684, 2020. 1, 4, 5

[13] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *CVPR*, pages 10684–10695, 2022. 1, 2

[14] Joonghyuk Shin, Minguk Kang, and Jaesik Park. Fill-up: Balancing long-tailed data with generative models. *arXiv preprint arXiv:2306.07200*, 2023. 1, 5

[15] Zeyu Wang, Klint Qinami, Ioannis Christos Karakozis, Kyle Genova, Prem Nair, Kenji Hata, and Olga Russakovsky. Towards fairness in visual recognition: Effective strategies for bias mitigation. In *CVPR*, pages 8919–8928, 2020. 1

[16] Moon Ye-Bin, Nam Hyeon-Woo, Wonseok Choi, Nayeong Kim, Suha Kwak, and Tae-Hyun Oh. Ex-

AI-Generated

Figure 7. Additional generations across prompts and seeds.

ploiting synthetic data for data imbalance problems: Baselines from a data perspective. *arXiv preprint arXiv:2308.00994*, 2023. 1, 5

[17] Fengda Zhang, Qianpei He, Kun Kuang, Jiashuo Liu, Long Chen, Chao Wu, Jun Xiao, and Hanwang Zhang. Distributionally generative augmentation for fair facial attribute classification. In *CVPR*, pages 22797–22808, 2024. 1, 4, 5

[18] Xiao Zhou, Yong Lin, Renjie Pi, Weizhong Zhang, Renzhe Xu, Peng Cui, and Tong Zhang. Model agnostic sample reweighting for out-of-distribution learning. In *ICML*, pages 27203–27221, 2022. 1, 4, 5